ព្រះរាជាណាចក្រកម្ពុជា
ជាតិ សាសនា ព្រះមហាក្សត្រ

សាកលវិទ្យាល័យភ្នំពេញអន្តរជាតិ

PHONM PENH INTERNATIONAL UNIVERSITY

Topic: UrbanCart Big Data Analytics

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ឯកទេស៖ Big Data Analytics

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Executive Summary

UrbanCart's cleaned completed-order transactions generated net revenue across six categories. Books & Media was the strongest category at $2,288,541.26, while the leading customer spent $145,169.05. The strongest RFM segment was Champions, contributing $2,455,158.51. The supplier reconciliation identified 12 supplier-only products and 45 database-only products.

The Normal Equation regression achieved R-squared 0.680. It projected revenue of $334,584.22 and $339,289.52 for the next two months. The review-rating/repeat-purchase correlation was 0.035, indicating that ratings alone provide little explanatory power for customer loyalty.

Top five findings

- Books & Media generated the highest category revenue.
- Champions customers generated the largest RFM-segment revenue.
- Beauty & Health had the highest return-unit rate at 2.45%.
- The supplier catalog contained 12 supplier-only SKUs and the database contained 45 database-only products.
- Rating and repeat-purchase correlation was weak (0.035).

Priority recommendations

- Protect availability for the strongest categories and highest-revenue products.
- Target Champions and high-value churn-risk customers with differentiated retention campaigns.
- Investigate product quality and content accuracy in high-return categories.
- Create a governed product-master process to reconcile internal and supplier catalogs.
- Use the revenue forecast as a planning range rather than a guaranteed point estimate.

1. Introduction and Business Context

UrbanCart is a fictional online retailer operating across fifteen cities and six product categories. Leadership requires an evidence-based view of customer value, product profitability, demand, and data quality. The project builds a complete pipeline from raw operational data and messy external files to reusable tables, NumPy models, charts, and management recommendations.

1.1 Business questions

13. Who are UrbanCart's most valuable customers, and what indicates loyalty or churn?
14. Which products and categories remain profitable after discounts, costs, and returns, and what is the near-term demand outlook?
15. Can the marketing and supplier data be trusted, and what controls are needed to improve it?

1.2 Analytical approach

Phase 1 extracts decision-ready results with SQL. Phase 2 cleans and reconciles data with pandas. Phase 3 demonstrates underlying numerical methods directly with NumPy. Phase 4 converts model output into eight business insights, while Phase 5 packages the workflow so it can be rerun with one command.

2. Data Description

The operational SQLite database contains customers, products, orders, order_items, reviews, and web_sessions. Two external spreadsheets represent a legacy marketing export and a supplier catalog. The source data intentionally contains missing values, duplicate or near-duplicate records, inconsistent date formats, invalid ratings, extreme prices, and partially overlapping product coverage.

Dataset	Purpose	Clean rows
customers	Customer identity and demographics	2500
products	Product master and price/cost	312
orders	Order headers and payment status	9000
order_items	Product-level transactions and returns	20176
reviews	Ratings and review text	4000
web_sessions	Engagement by device	12000

2.1 Known data limitations

- Missing age, city, gender, and review text require field-specific policies.
- Returns are represented by negative quantities rather than a separate returns table.
- Supplier and database products have incomplete overlap.
- Forecasting is based on a limited historical window and assumes recent relationships continue.

3. SQL Extraction and Methodology

All ten SQL queries are stored in queries.sql and are run directly against ecommerce.db. Results are loaded into pandas and exported to outputs/tables.

3.1 Query 1 - Category revenue and average order value

category	total_revenue	total_orders	average_order_value
Books & Media	2,288,541.26	1649	1,387.84
Beauty & Health	1,781,007.78	1637	1,087.97
Apparel	967,057.00	1629	593.65
Sports & Outdoors	916,247.78	1579	580.27
Electronics	900,204.48	1615	557.40
Home & Kitchen	781,563.91	1660	470.82

Interpretation

This query identifies the revenue concentration across categories using completed orders and the correct fractional-discount formula.

3.2 Query 2 - Top 20 customers

customer_id	name	city	signup_date	total_orders	lifetime_spending
611	Tara Holmes	Sydney	2021-10-15	3	145,169.05
689	Misty Mcguire	Toronto	2020-08-25	2	107,213.91
1259	Matthew Perez	Los Angeles	2023-02-25	2	94,519.39
1217	Virginia Hernandez	Paris	2022-07-07	4	77,530.81
528	Brent Hopkins	Vancouver	2023-10-12	6	72,405.36
2445	Virginia Rose	Bangkok	2023-11-26	3	72,021.09
1993	Chelsea Austin	Sydney	2020-04-27	1	70,254.39
1347	Evan Davies	Paris	2020-05-28	2	70,236.44
1533	Lauren Anthony	Vancouver	2023-10-03	5	59,602.71
1180	Tammy Booth	Mumbai	2022-03-30	5	42,127.63
634	Dr. Samuel Holmes DDS	New York	2019-04-09	6	41,632.49
1316	Susan Sanchez	Bangkok	2024-04-25	5	40,678.11

Interpretation

This customer-ranking query combines customer, order, and order-item tables. It supports high-value customer retention and service prioritization.

3.3 Query 3 - Month-over-month revenue

revenue_month	revenue	previous_month_revenue	revenue_change	percentage_change
2023-01	141,785.11	nan	nan	nan
2023-02	135,209.46	141,785.11	-6,575.66	-4.64
2023-03	172,275.15	135,209.46	37,065.70	27.41
2023-04	106,562.90	172,275.15	-65,712.26	-38.14
2023-05	312,271.28	106,562.90	205,708.39	193.04
2023-06	226,846.85	312,271.28	-85,424.44	-27.36
2023-07	200,398.05	226,846.85	-26,448.80	-11.66
2023-08	164,611.43	200,398.05	-35,786.62	-17.86
2023-09	321,632.97	164,611.43	157,021.54	95.39
2023-10	197,016.29	321,632.97	-124,616.68	-38.74
2023-11	321,623.83	197,016.29	124,607.54	63.25
2023-12	244,383.27	321,623.83	-77,240.56	-24.02

Interpretation

The LAG window function compares each month with the previous month over the latest 24-month period.

3.4 Query 4 - Return rates by category

category	item_rows	return_rows	returned_units	total_units	return_row_rate_percent	return_unit_rate_percent
Beauty & Health	3354	106	106	4328	3.16	2.45
Electronics	3401	106	106	4425	3.12	2.40
Apparel	3372	102	102	4355	3.02	2.34
Books & Media	3455	93	93	4461	2.69	2.08
Home & Kitchen	3468	91	91	4439	2.62	2.05
Sports & Outdoors	3312	85	85	4252	2.57	2.00

Interpretation

Negative quantities are treated as returns. The unit-based rate is more informative than simply counting returned rows.

3.5 Query 5 - Latest three-quarter customers

customer_id	name	city	signup_date	active_required_quarters
73	Alicia Gilmore	Los Angeles	2023-01-14	3
2497	Anthony Velasquez	Paris	2023-09-28	3
2189	Bradley Brady	Toronto	2024-04-01	3
1036	Brenda Gray	New York	2024-06-04	3
528	Brent Hopkins	Vancouver	2023-10-12	3
2164	Brittany Barnett	Berlin	2023-10-20	3
405	Cassidy Horne	Sydney	2023-04-05	3
1494	Christine Watson	Chicago	2023-08-27	3
2150	Cindy Cole	Vancouver	2022-10-20	3
2069	Corey Jones	Melbourne	2019-07-27	3
2419	Crystal Brown	Vancouver	2024-05-13	3
1772	Danielle Kelly	Mumbai	2023-07-02	3

Interpretation

Customers in this output purchased in every one of the latest three calendar quarters, indicating sustained engagement.

3.6 Query 6 - Top-rated reviewed products

product_id	name	category	total_reviews	average_rating
200	White Fitnes	Sports & Outdoors	15	4.60
19	Improve Smartphone	Electronics	15	4.53
283	Strategy Children	Books & Media	15	4.53
191	Yeah Camping	Sports & Outdoors	15	4.53
21	Right Camera	Electronics	16	4.50
182	Answer Fitnes	Sports & Outdoors	15	4.40
66	Wide Cookware	Home & Kitchen	15	4.40
122	Attorney Men	Apparel	17	4.35
208	Series Skincare	Beauty & Health	18	4.33
154	Attack Cycling	Sports & Outdoors	16	4.31

Interpretation

Only valid ratings from 1 to 5 are included, and products require at least 15 reviews to reduce small-sample distortion.

3.7 Query 7 - Purchased-customer session behavior

device	total_sessions	purchasing_customers	average_duration_minutes	average_pages_viewed
desktop	3494	1776	6.25	4.68
mobile	3458	1719	6.20	4.65
tablet	3569	1761	6.09	4.74

Interpretation

EXISTS restricts sessions to customers with at least one completed purchase, linking engagement with conversion.

3.8 Query 8 - Product revenue ranking

category	product_id	name	net_revenue	revenue_rank
Apparel	139	About Men	46,240.50	1
Apparel	104	Major Men	45,950.88	2
Apparel	136	Watch Footwear	45,845.05	3
Apparel	131	Spring Kid	43,443.58	4
Apparel	107	Appear Women	40,099.07	5
Apparel	142	Card Footwear	39,954.72	6
Apparel	123	Anything Footwear	39,229.85	7
Apparel	116	Ahead Men	36,435.13	8
Apparel	117	White Men	29,974.03	9
Apparel	125	Increase Women	29,440.15	10
Apparel	135	Road Women	29,386.78	11
Apparel	134	Democrat Kid	28,741.96	12

Interpretation

DENSE_RANK compares product revenue within category and provides a basis for merchandising and replenishment.

3.9 Query 9 - Country payment mix

country	payment_method	order_count	payment_method_share_percent
Australia	gift_card	165	22.27
Australia	credit_card	162	21.86
Australia	debit_card	141	19.03
Australia	paypal	137	18.49
Australia	bank_transfer	136	18.35
Canada	credit_card	152	21.41
Canada	gift_card	149	20.99
Canada	paypal	146	20.56
Canada	debit_card	140	19.72
Canada	bank_transfer	123	17.32
France	paypal	73	22.32
France	debit_card	67	20.49

Interpretation

Payment shares are calculated within each country, supporting local checkout and provider decisions.

3.10 Query 10 - High-value churn risk

customer_id	name	city	country	last_order_date	completed_orders	lifetime_revenue	days_since_last_order	customer_status
689	Misty Mcguire	Toronto	Canada	2023-09-09	2	107,213.91	479	High-value churn risk
1259	Matthew Perez	Los Angeles	USA	2024-07-11	2	94,519.39	173	High-value churn risk
1217	Virginia Hernandez	Paris	France	2024-08-12	4	77,530.81	141	High-value churn risk
1993	Chelsea Austin	Sydney	Australia	2022-03-23	1	70,254.39	1014	High-value churn risk
1347	Evan Davies	Paris	France	2023-08-23	2	70,236.44	496	High-value churn risk
1180	Tammy Booth	Mumbai	India	2024-08-14	5	42,127.63	139	High-value churn risk
634	Dr. Samuel Holmes DDS	New York	USA	2024-08-26	6	41,632.49	127	High-value churn risk
1357	Ashley Cordova	Singapore	Singapore	2024-01-31	4	39,417.30	335	High-value churn risk
1999	Kayla Fisher	Chicago	USA	2024-02-11	6	38,890.85	324	High-value churn risk
123	Mary Grimes	Chicago	USA	2024-04-05	4	38,698.77	270	High-value churn risk
2287	Norma Smith	Manchester	UK	2024-05-20	4	38,640.02	225	High-value churn risk
298	Austin Wheeler	Mumbai	India	2024-01-29	2	38,003.02	337	High-value churn risk

Interpretation

The custom leadership query combines lifetime revenue and purchase recency to prioritize retention outreach.

4. Data Cleaning and Integration

Cleaning decisions were documented and audit tables were exported. The pipeline avoids global fillna(0), does not delete legitimate returns, and preserves original price values when outlier treatment is applied.

4.1 Legacy customer dates and duplicates

metric	value
Original rows	1427
Blank/junk rows removed	2
Exact duplicates found	0
Rows after near-duplicate resolution	1377
Unparsed signup dates	0
Rows with missing email	56

Rule and business impact

The legacy file was reduced from 1,427 rows to 1,377 after blank/junk removal and normalized-email near-duplicate resolution. All signup dates were parsed.

4.2 Missing-value policies

field	policy	affected_rows
age	Median imputation (46.0) with missing flag	139
city	Unknown with missing flag	63
gender	Unknown with missing flag	115
review_text	No written review with missing flag	838
rating	Outside 1–5 converted to missing and audited	41

Rule and business impact

Age was median-imputed with a flag. City and gender were represented as Unknown. Missing review text remained analytically distinguishable, and invalid ratings were excluded.

4.3 Product price outliers

product_id	name	category	subcategory	unit_price	cost	unit_price_original	unit_price_is_outlier
245	Throughout Personal Care	Beauty & Health	Personal Care	11,506.50	104.23	11,506.50	True
250	Discover Haircare	Beauty & Health	Haircare	8,284.00	112.79	8,284.00	True
270	Environmental Non-Fiction	Books & Media	Non-Fiction	34,872.00	252.13	34,872.00	True

Rule and business impact

IQR thresholds detected extreme product prices. Original values and outlier flags were preserved while analytical prices were capped.

4.4 Duplicate order items and returns

order_item_id	order_id	product_id	quantity	unit_price	discount
22.00	8.00	240.00	1.00	555.47	0.00
49.00	18.00	47.00	1.00	513.39	0.00
243.00	103.00	255.00	1.00	249.20	0.00
557.00	233.00	234.00	2.00	486.72	0.00
825.00	356.00	50.00	1.00	240.10	0.00
930.00	402.00	54.00	2.00	679.71	0.20
1,054.00	458.00	142.00	1.00	717.32	0.00
1,176.00	512.00	245.00	1.00	11,506.50	0.10
1,301.00	569.00	274.00	3.00	528.78	0.00
1,499.00	664.00	3.00	2.00	978.72	0.10
1,504.00	665.00	154.00	1.00	115.68	0.20
1,540.00	677.00	284.00	1.00	228.90	0.00
1,717.00	770.00	184.00	1.00	278.40	0.05
1,761.00	789.00	85.00	1.00	530.70	0.05
1,932.00	865.00	130.00	3.00	454.11	0.05

Rule and business impact

Duplicate business transactions were removed without using the primary-key field. Negative quantities were retained as returns in net-revenue calculations.

4.5 Product catalog reconciliation

metric	value
Internal products	300
Supplier products	267
Matched	255
Supplier only	12
Database only	45

Rule and business impact

An outer merge explicitly reported matched, supplier-only, and database-only products. The operational database remained authoritative for products already used in orders.

4.6 Pivot table and time series

A category-by-month net-revenue pivot matrix was created and validated against transaction totals. A weekly time series measures active customers, completed orders, revenue per active customer, growth, and four-week moving averages. These datasets directly support Phase 4 trend charts.

category	2022-01-01 00:00:00	2022-02-01 00:00:00	2022-03-01 00:00:00	2022-04-01 00:00:00	2022-05-01 00:00:00	2022-06-01 00:00:00	2022-07-01 00:00:00
Apparel	13,435.77	11,201.88	13,153.09	17,600.82	19,910.74	17,416.09	11,459.81
Beauty & Health	44,463.63	36,146.92	8,114.15	32,643.76	38,765.76	32,297.28	50,848.59
Books & Media	15,161.12	13,794.73	83,262.29	11,741.47	15,833.36	12,814.93	78,250.49
Electronics	14,382.26	13,144.08	12,960.27	17,476.00	23,032.55	12,498.16	13,999.06
Home & Kitchen	14,818.81	11,599.69	16,170.24	4,910.22	10,054.60	14,004.21	13,148.64
Sports & Outdoors	16,727.53	12,635.17	12,675.21	9,927.15	22,207.70	15,030.78	11,874.10

5. NumPy Analytical Methods

The four required routines use raw NumPy arrays for scoring, similarity, regression, and simulation.

5.1 RFM segmentation

Formula

RFM score = R + F + M

Method and validation

Recency, frequency, and monetary values were bucketed with NumPy percentiles. Recency is reverse-scored because lower values are better. The composite score supports customer segmentation.

rfm_segment	customers	average_recency	average_frequency	total_revenue
Champions	329	67.97	4.02	2,455,158.51
Potential	894	162.12	2.06	2,413,432.60
Hibernating	682	601.07	1.41	1,540,737.23
At Risk	169	481.43	3.47	964,519.68
Loyal	117	68.97	3.28	195,639.96

5.2 Cosine-similarity recommendation

Formula

cosine(x,y) = (x dot y) / (||x|| ||y||)

Method and validation

Product vectors were normalized with NumPy norms and multiplied with a dot product. No sklearn cosine_similarity shortcut was used.

product_id	product_name	recommended_product_id	recommended_product_name	similarity	rank
1	Black Camera	184	Important Cycling	0.11	1
1	Black Camera	244	Thus Supplement	0.11	2
1	Black Camera	90	Group Furniture	0.11	3
1	Black Camera	97	Black Appliance	0.11	4
1	Black Camera	293	Almost Non-Fiction	0.11	5
2	Maintain Laptop	132	Set Men	0.11	1
2	Maintain Laptop	95	They Appliance	0.10	2
2	Maintain Laptop	149	Move Kid	0.09	3
2	Maintain Laptop	257	Minute Children	0.09	4
2	Maintain Laptop	164	Great Team Sport	0.09	5
3	Start Accessorie	208	Series Skincare	0.13	1
3	Start Accessorie	68	Bar Appliance	0.13	2
3	Start Accessorie	194	Quickly Fitnes	0.09	3
3	Start Accessorie	34	Television Laptop	0.09	4
3	Start Accessorie	71	Green Furniture	0.09	5

5.3 Regression through the Normal Equation

Formula

beta = (X^T X)^-1 X^T y

Method and validation

A linear model was fitted with the pseudo-inverse form of the Normal Equation. R-squared, RMSE, and MAE were computed manually.

metric	value
R-squared	0.68
RMSE	51,892.62
MAE	41,098.31

5.4 Monte Carlo simulation

Formula

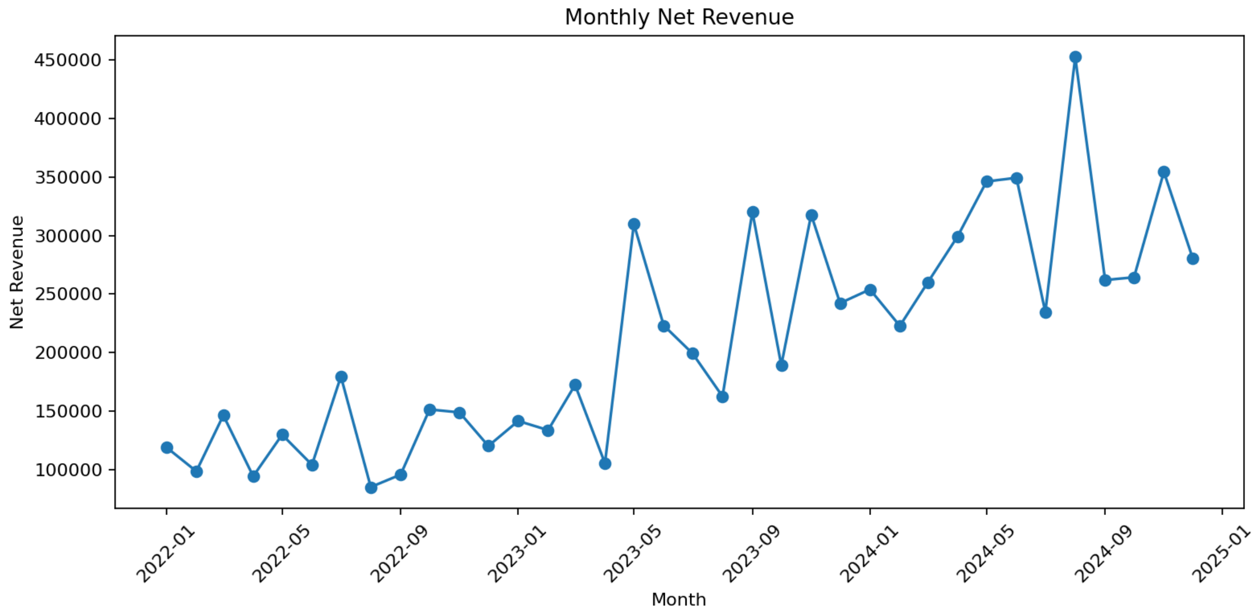
Revenue(t+1) = Revenue(t) x (1 + sampled growth)

Method and validation

Ten thousand revenue paths sampled empirical monthly growth rates. The 5th and 95th percentiles form a 90% uncertainty interval.

month_ahead	mean_revenue	median_revenue	p05	p95	probability_below_current
1.00	313,324.25	282,998.67	162,331.08	552,418.53	0.48
2.00	351,168.75	286,481.30	125,518.47	803,794.01	0.48

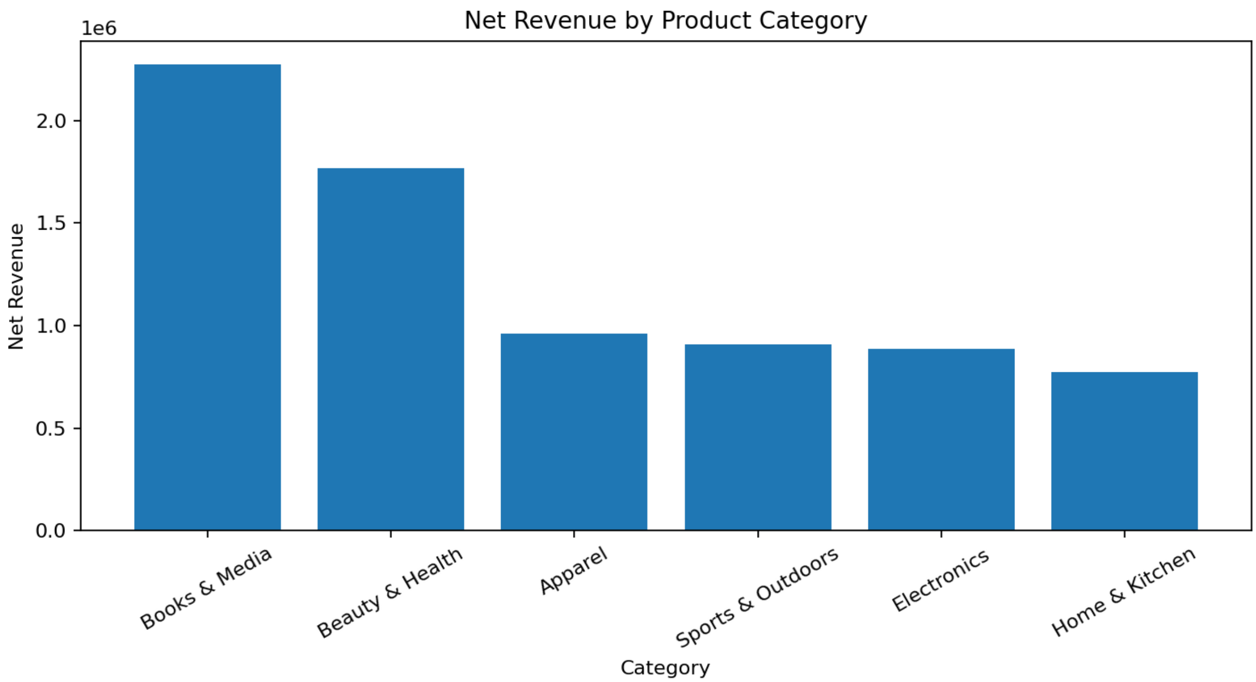
6.1 Monthly revenue trend



Business interpretation

The latest 24-month SQL results show substantial month-to-month volatility. The highest displayed month was 2024-08 at $454,886.62. Planning should therefore use ranges and moving averages.

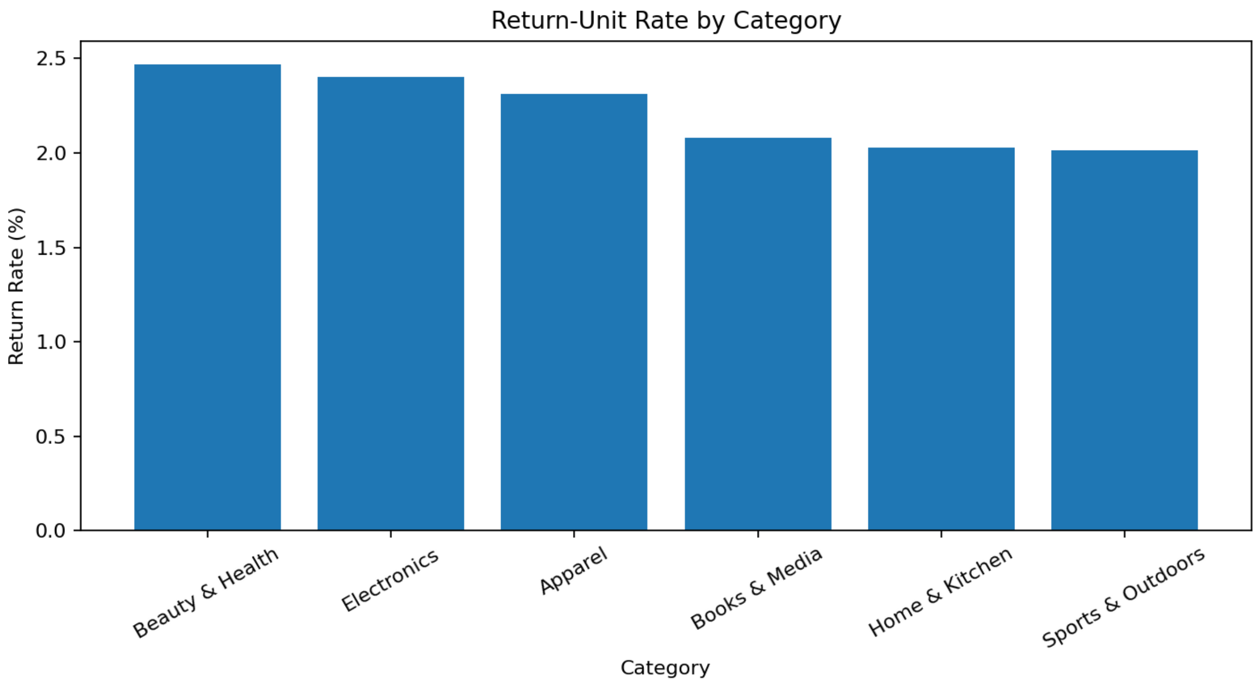
6.2 Category revenue performance



Business interpretation

Books & Media generated the highest category revenue at $2,288,541.26. Inventory protection and marketing support should prioritize this category while validating margin.

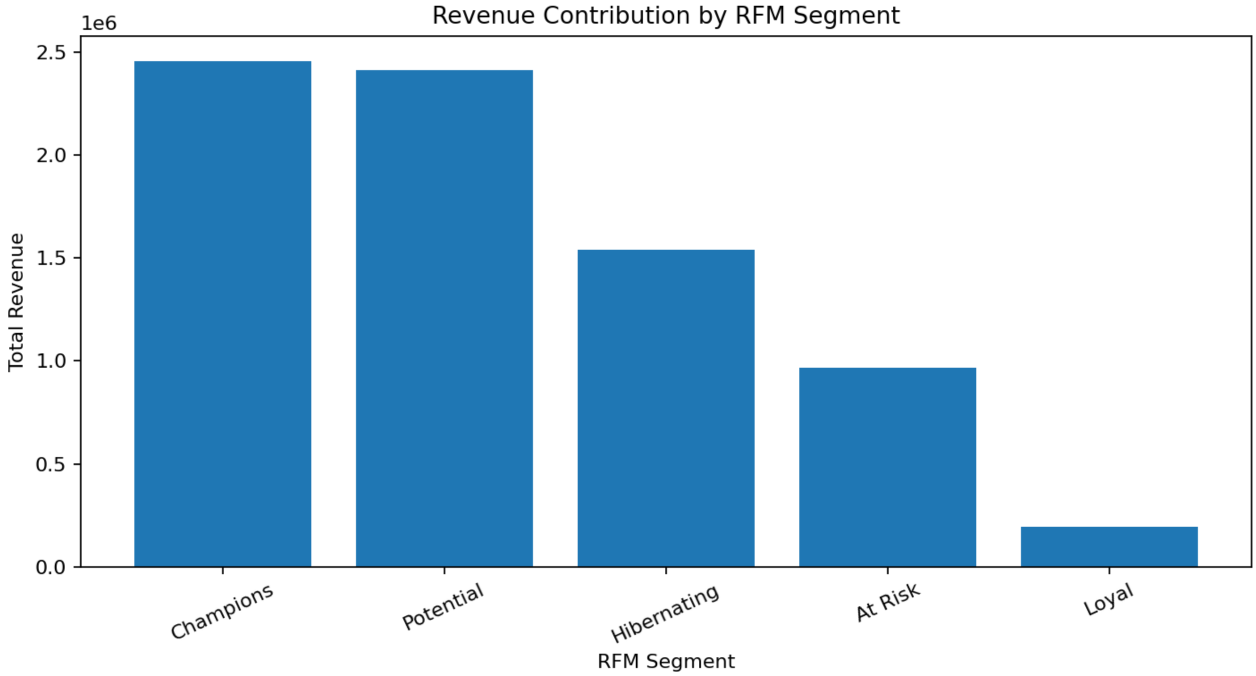
6.3 Category return rate



Business interpretation

Beauty & Health had the highest return-unit rate at 2.45%. Management should inspect product quality, descriptions, packaging, and supplier issues.

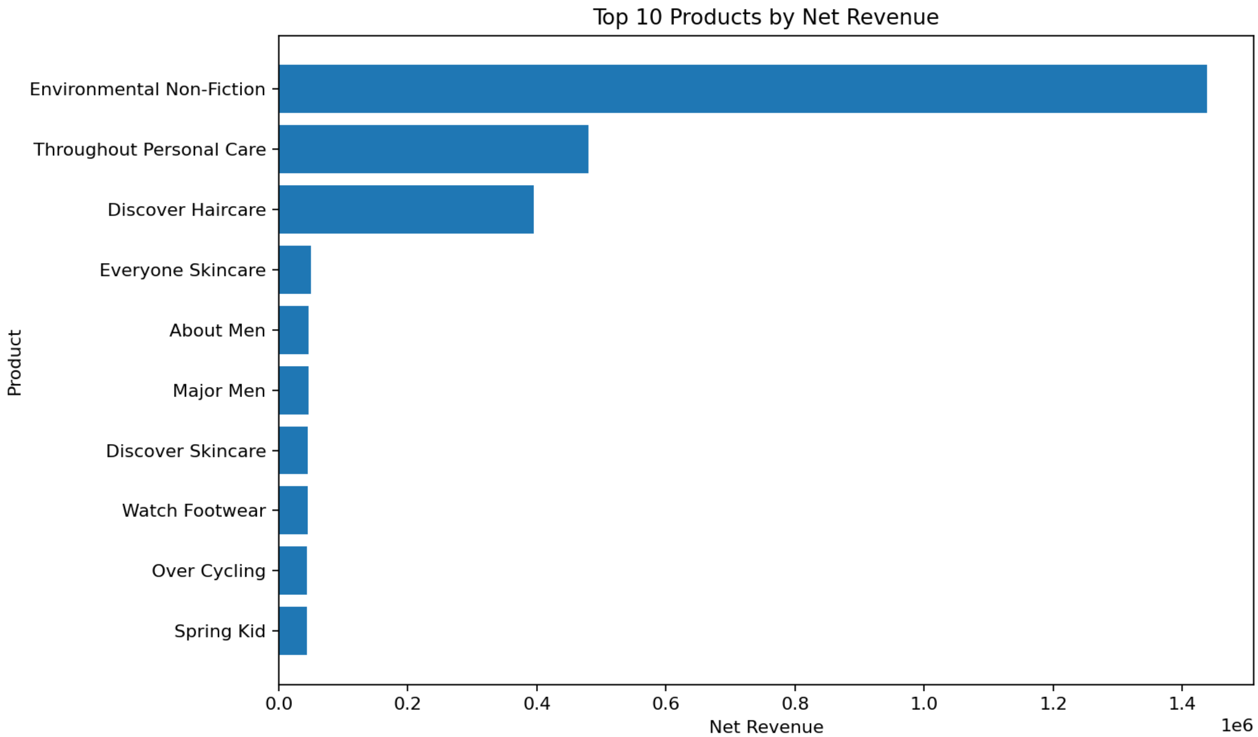
6.4 RFM segment revenue



Business interpretation

Champions generated the largest segment revenue at $2,455,158.51. Retention investment should reflect both revenue contribution and churn risk.

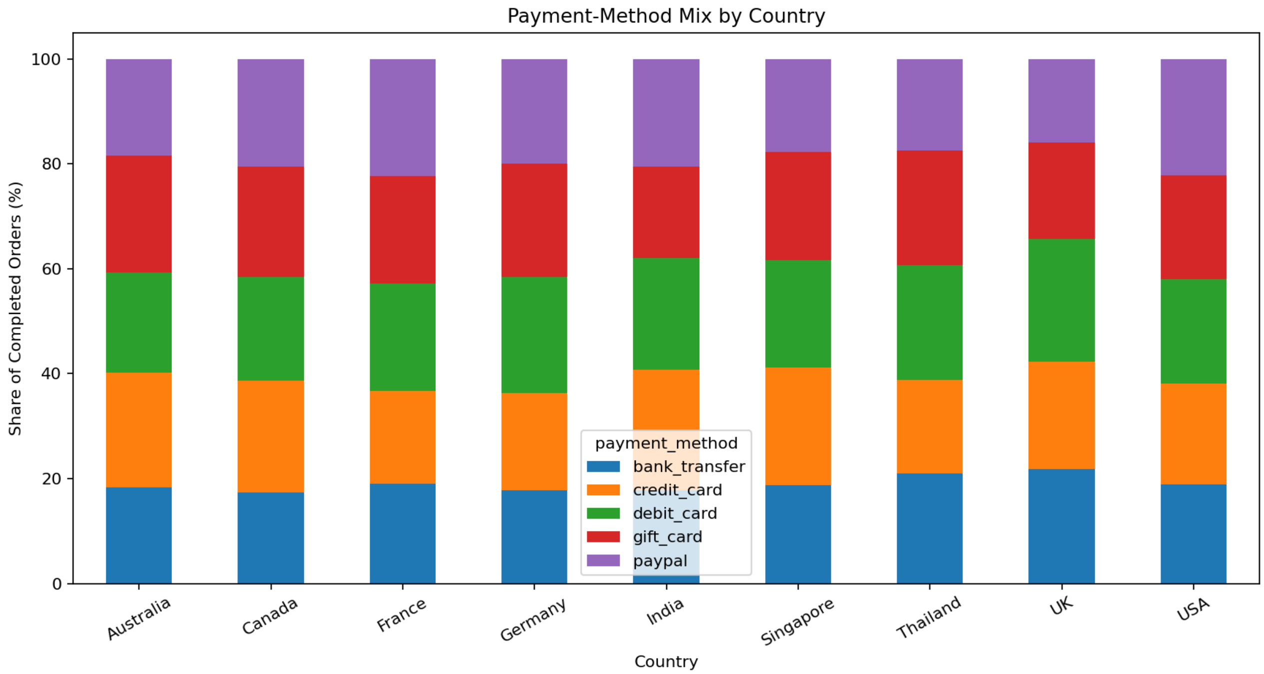
6.5 Top products



Business interpretation

Revenue is concentrated in a relatively small set of products. Replenishment and recommendation placement should protect these items, while return and margin performance are monitored.

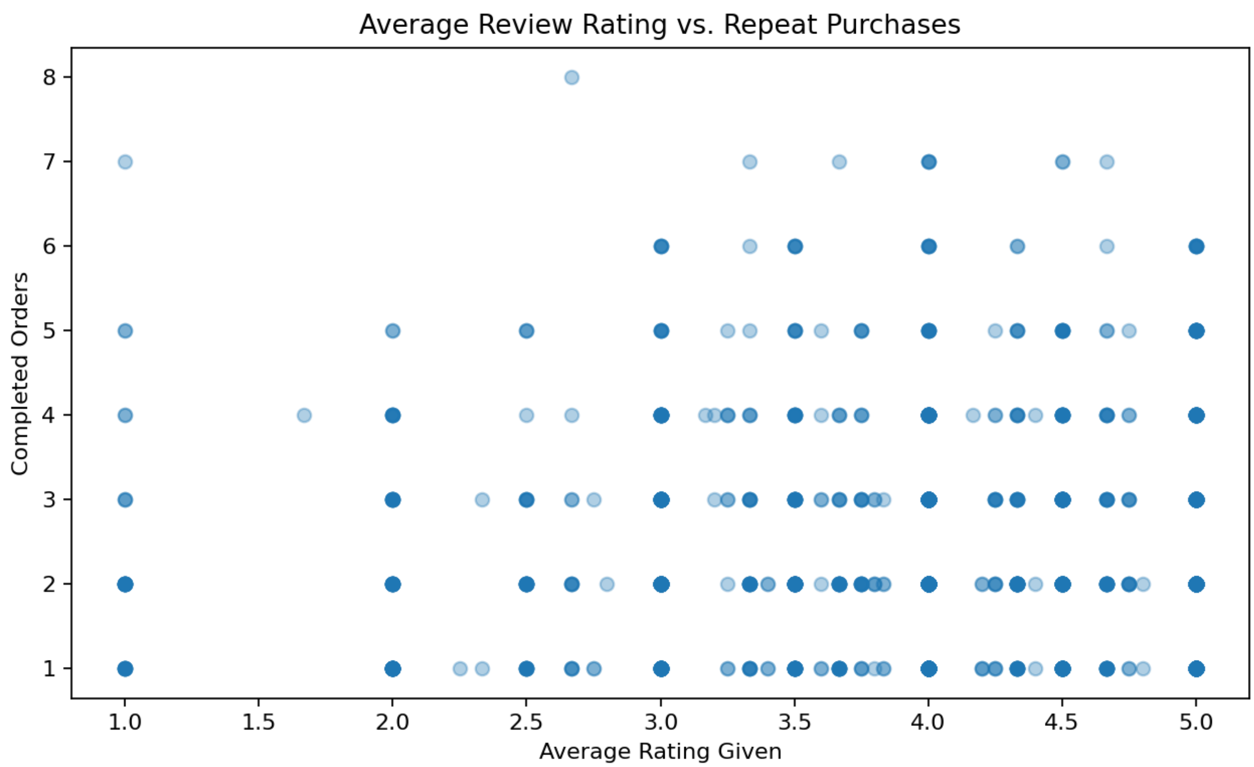
6.6 Payment method by country



Business interpretation

Payment preferences vary across countries. Localized checkout ordering and provider coverage can reduce abandonment and improve conversion.

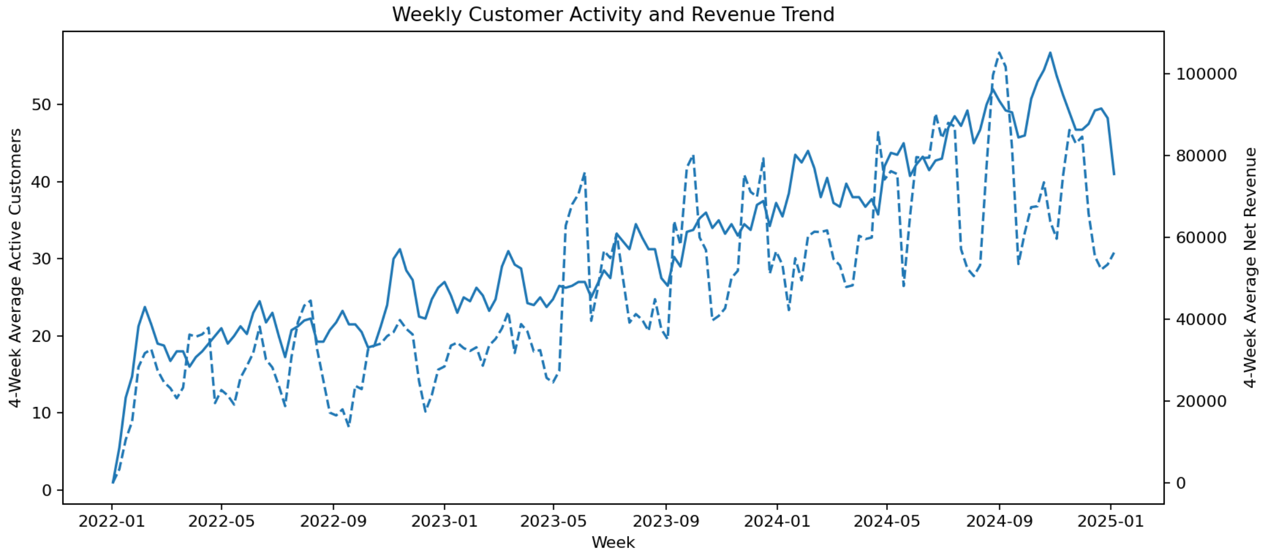
6.7 Ratings and repeat purchases



Business interpretation

The Pearson correlation between average rating given and completed orders is 0.035. This weak relationship means loyalty models should include recency, frequency, value, and engagement.

6.8 Weekly customer activity and revenue



Business interpretation

Four-week moving averages reveal whether engagement and revenue move together. Divergence can indicate changing order value, discounting, product mix, or customer quality.

7. Limitations and Threats to Validity

- The data is fictional and may not capture every operational process found in a real retailer.
- Negative quantity is assumed to represent a return; no separate return reason or refund table is available.
- Outlier capping reduces influence but may alter legitimately premium product prices.
- Median age imputation reduces missingness but understates uncertainty and distribution variability.
- The regression model is descriptive and depends on the stability of historical relationships.
- Monte Carlo paths sample historical growth behavior and do not model future structural changes.
- Supplier-only products lack historical orders, preventing direct demand validation.
- Correlation does not establish causation between review behavior and repeat purchasing.

8. Recommendations and Next Steps

1. Protect high-value demand

Use category and product rankings to prioritize replenishment, but add margin and return-rate controls before committing inventory.

2. Establish segmented retention

Build separate campaigns for Champions, Loyal, At Risk, and high-value churn-risk customers.

3. Reduce return drivers

Review high-return categories at SKU and supplier level, including quality, sizing, descriptions, and packaging.

4. Govern the product master

Create ownership, matching rules, and exception queues for supplier-only and database-only SKUs.

5. Operationalize forecast ranges

Use regression and Monte Carlo outputs as planning ranges with monthly recalibration.

9. Reusable Pipeline

Phase 5 packages Phases 1-4 into a one-command driver. Running python main.py validates the raw inputs, executes analysis.ipynb, and regenerates clean datasets, SQL tables, NumPy model results, and charts. A Makefile provides install, run, and clean targets.

Commands

```
python -m pip install -r requirements.txt

python main.py

make run
```

GitHub repository structure

```
UrbanCart_Complete_Project/
├── analysis.ipynb
├── queries.sql
├── main.py
├── Makefile
├── requirements.txt
├── README.md
├── src/pipeline.py
├── data/raw/
├── data/processed/
└── outputs/{tables,model_results,charts}/
```

Appendix A - SQL Listing

SQL listing - Part 1

```
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==================================================
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-- UrbanCart Big Data Analytics
-- Phase 1: SQL Data Extraction
-- Database: SQLite (ecommerce.db)
--
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=========


-- Query 1: Calculate net revenue, order count, and
average order
-- value by product category after accounting for
discounts.
SELECT
    p.category,
    ROUND(
        SUM(
            oi.quantity
            * oi.unit_price
            * (1 - COALESCE(oi.discount, 0))
        ),
        2
    ) AS total_revenue,
    COUNT(DISTINCT o.order_id) AS total_orders,
    ROUND(
        SUM(
            oi.quantity
```

```
                * oi.unit_price
                * (1 - COALESCE(oi.discount, 0))
            ) / NULLIF(COUNT(DISTINCT o.order_id), 0),
            2
        ) AS average_order_value
FROM order_items AS oi
JOIN products AS p
    ON oi.product_id = p.product_id
JOIN orders AS o
    ON oi.order_id = o.order_id
WHERE LOWER(o.status) = 'completed'
GROUP BY p.category
ORDER BY total_revenue DESC;


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-- Query 2: Identify the top 20 customers by
lifetime spending,
-- including their city and signup date.
SELECT
    c.customer_id,
    c.name,
    c.city,
    c.signup_date,
    COUNT(DISTINCT o.order_id) AS total_orders,
    ROUND(
        SUM(
            oi.quantity
            * oi.unit_price
```

```
                * (1 - COALESCE(oi.discount, 0))
            ),
            2
        ) AS lifetime_spending
FROM customers AS c
JOIN orders AS o
    ON c.customer_id = o.customer_id
JOIN order_items AS oi
    ON o.order_id = oi.order_id
WHERE LOWER(o.status) = 'completed'
GROUP BY
    c.customer_id,
    c.name,
    c.city,
    c.signup_date
ORDER BY lifetime_spending DESC
LIMIT 20;


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-- Query 3: Show the month-over-month revenue trend
for the most
-- recent 24 months, including previous revenue and
percentage change.
WITH monthly_revenue AS (
    SELECT
        STRFTIME('%Y-%m', o.order_date) AS
revenue_month,
        ROUND(
```

```
            SUM(
                oi.quantity
                * oi.unit_price
                * (1 - COALESCE(oi.discount, 0))
            ),
            2
        ) AS revenue
    FROM orders AS o
    JOIN order_items AS oi
        ON o.order_id = oi.order_id
    WHERE LOWER(o.status) = 'completed'
    GROUP BY STRFTIME('%Y-%m', o.order_date)
),
recent_24_months AS (
    SELECT
        revenue_month,
        revenue
    FROM monthly_revenue
    ORDER BY revenue_month DESC
    LIMIT 24
),
monthly_comparison AS (
    SELECT
        revenue_month,
        revenue,
        LAG(revenue) OVER (
            ORDER BY revenue_month
        ) AS previous_month_revenue
    FROM recent_24_months
)
SELECT
    revenue_month,
    revenue,
```

```
    previous_month_revenue,
    ROUND(
        revenue - previous_month_revenue,
        2
    ) AS revenue_change,
    ROUND(
        100.0 * (revenue - previous_month_revenue)
        / NULLIF(previous_month_revenue, 0),
        2
    ) AS percentage_change
FROM monthly_comparison
ORDER BY revenue_month;


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-- Query 4: Calculate the return rate for each
category. Negative
-- quantities represent returned units.
WITH category_item_summary AS (
    SELECT
        p.category,
        COUNT(*) AS total_item_rows,
        SUM(
            CASE
                WHEN oi.quantity < 0 THEN 1
                ELSE 0
            END
        ) AS returned_item_rows,
        SUM(ABS(oi.quantity)) AS total_units,
```

```
        SUM(
            CASE
                WHEN oi.quantity < 0 THEN
ABS(oi.quantity)
                ELSE 0
            END
        ) AS returned_units
    FROM order_items AS oi
    JOIN products AS p
        ON oi.product_id = p.product_id
    GROUP BY p.category
)
SELECT
    category,
    total_item_rows,
    returned_item_rows,
    returned_units,
    total_units,
    ROUND(
        100.0 * returned_item_rows
        / NULLIF(total_item_rows, 0),
        2
    ) AS return_row_rate_percent,
    ROUND(
        100.0 * returned_units
        / NULLIF(total_units, 0),
        2
    ) AS return_unit_rate_percent
FROM category_item_summary
ORDER BY return_unit_rate_percent DESC;


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-- Query 5: Find customers who placed at least one
completed order
-- in every one of the latest three calendar
quarters in the data.
WITH latest_date AS (
    SELECT
```

SQL listing - Part 2

```
        MAX(DATE(order_date)) AS maximum_order_date
    FROM orders
),
latest_quarter AS (
    SELECT
        DATE(
            STRFTIME('%Y', maximum_order_date)
            || '-'
            || PRINTF(
                '%02d',
                (
                    (
                        CAST(STRFTIME('%m',
maximum_order_date) AS INTEGER)
                        - 1
                    ) / 3
                ) * 3 + 1
            )
            || '-01'
        ) AS current_quarter_start
    FROM latest_date
),
required_quarters AS (
    SELECT current_quarter_start AS quarter_start
    FROM latest_quarter

    UNION ALL

    SELECT DATE(current_quarter_start, '-3 months')
    FROM latest_quarter
```

```
    UNION ALL

    SELECT DATE(current_quarter_start, '-6 months')
    FROM latest_quarter
),
customer_quarters AS (
    SELECT DISTINCT
        o.customer_id,
        DATE(
            STRFTIME('%Y', o.order_date)
            || '-'
            || PRINTF(
                '%02d',
                (
                    (
                        CAST(STRFTIME('%m',
o.order_date) AS INTEGER)
                        - 1
                    ) / 3
                ) * 3 + 1
            )
            || '-01'
        ) AS quarter_start
    FROM orders AS o
    WHERE LOWER(o.status) = 'completed'
)
SELECT
    c.customer_id,
    c.name,
    c.city,
    c.signup_date,
    COUNT(DISTINCT cq.quarter_start) AS
active_required_quarters
```

```
FROM customers AS c
JOIN customer_quarters AS cq
    ON c.customer_id = cq.customer_id
JOIN required_quarters AS rq
    ON cq.quarter_start = rq.quarter_start
GROUP BY
    c.customer_id,
    c.name,
    c.city,
    c.signup_date
HAVING COUNT(DISTINCT cq.quarter_start) = 3
ORDER BY c.name;


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-- Query 6: Return the 10 products with the highest
valid average
-- review rating among products with at least 15
valid reviews.
SELECT
    p.product_id,
    p.name,
    p.category,
    COUNT(r.review_id) AS total_reviews,
    ROUND(AVG(r.rating), 2) AS average_rating
FROM products AS p
JOIN reviews AS r
    ON p.product_id = r.product_id
WHERE r.rating BETWEEN 1 AND 5
```

```
GROUP BY
    p.product_id,
    p.name,
    p.category
HAVING COUNT(r.review_id) >= 15
ORDER BY
    average_rating DESC,
    total_reviews DESC,
    p.name
LIMIT 10;


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-- Query 7: Calculate average session duration and
pages viewed by
-- device for customers who made at least one
completed purchase.
SELECT
    ws.device,
    COUNT(*) AS total_sessions,
    COUNT(DISTINCT ws.customer_id) AS
purchasing_customers,
    ROUND(AVG(ws.duration_minutes), 2) AS
average_duration_minutes,
    ROUND(AVG(ws.pages_viewed), 2) AS
average_pages_viewed
FROM web_sessions AS ws
WHERE EXISTS (
    SELECT 1
```

```
    FROM orders AS o
    WHERE o.customer_id = ws.customer_id
      AND LOWER(o.status) = 'completed'
)
GROUP BY ws.device
ORDER BY average_duration_minutes DESC;


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-- Query 8: Rank products by net revenue within
each product category.
WITH product_revenue AS (
    SELECT
        p.category,
        p.product_id,
        p.name,
        ROUND(
            SUM(
                oi.quantity
                * oi.unit_price
                * (1 - COALESCE(oi.discount, 0))
            ),
            2
        ) AS net_revenue
    FROM products AS p
    JOIN order_items AS oi
        ON p.product_id = oi.product_id
    JOIN orders AS o
        ON oi.order_id = o.order_id
```

```
    WHERE LOWER(o.status) = 'completed'
    GROUP BY
        p.category,
        p.product_id,
        p.name
)
SELECT
    category,
    product_id,
    name,
    net_revenue,
    DENSE_RANK() OVER (
        PARTITION BY category
        ORDER BY net_revenue DESC
    ) AS revenue_rank
FROM product_revenue
ORDER BY
    category,
    revenue_rank,
    name;


--
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-- Query 9: Show the payment-method mix by customer
country as both
-- order count and percentage share of orders
within each country.
WITH country_payment_orders AS (
    SELECT
```

```
        c.country,
        o.payment_method,
        COUNT(DISTINCT o.order_id) AS order_count
    FROM customers AS c
    JOIN orders AS o
        ON c.customer_id = o.customer_id
    WHERE LOWER(o.status) = 'completed'
    GROUP BY
        c.country,
        o.payment_method
)
SELECT
    country,
    payment_method,
    ord
```

SQL listing - Part 3

```
er_count,
    ROUND(
        100.0 * order_count
        / SUM(order_count) OVER (
            PARTITION BY country
        ),
        2
    ) AS payment_method_share_percent
FROM country_payment_orders
ORDER BY
    country,
    payment_method_share_percent DESC;


--
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-- Query 10: Leadership should identify high-value
customers who may
-- be at risk of churn so retention campaigns can
be prioritized.
WITH customer_activity AS (
    SELECT
        c.customer_id,
        c.name,
        c.city,
        c.country,
        MAX(DATE(o.order_date)) AS last_order_date,
        COUNT(DISTINCT o.order_id) AS
```

```
completed_orders,
        ROUND(
            SUM(
                oi.quantity
                * oi.unit_price
                * (1 - COALESCE(oi.discount, 0))
            ),
            2
        ) AS lifetime_revenue
    FROM customers AS c
    JOIN orders AS o
        ON c.customer_id = o.customer_id
    JOIN order_items AS oi
        ON o.order_id = oi.order_id
    WHERE LOWER(o.status) = 'completed'
    GROUP BY
        c.customer_id,
        c.name,
        c.city,
        c.country
),
dataset_reference_date AS (
    SELECT MAX(DATE(order_date)) AS
latest_order_date
    FROM orders
)
SELECT
    ca.customer_id,
    ca.name,
    ca.city,
    ca.country,
    ca.last_order_date,
    CAST(
```

```
        JULIANDAY(drd.latest_order_date)
        - JULIANDAY(ca.last_order_date)
        AS INTEGER
    ) AS days_since_last_order,
    ca.completed_orders,
    ca.lifetime_revenue,
    CASE
        WHEN ca.lifetime_revenue >= 5000
             AND (
                 JULIANDAY(drd.latest_order_date)
                 - JULIANDAY(ca.last_order_date)
             ) >= 90
        THEN 'High-value churn risk'

        WHEN ca.lifetime_revenue >= 5000
        THEN 'High-value active'

        WHEN (
             JULIANDAY(drd.latest_order_date)
             - JULIANDAY(ca.last_order_date)
             ) >= 90
        THEN 'Inactive'

        ELSE 'Active'
    END AS customer_status
FROM customer_activity AS ca
CROSS JOIN dataset_reference_date AS drd
ORDER BY
    CASE customer_status
        WHEN 'High-value churn risk' THEN 1
        WHEN 'High-value active' THEN 2
        WHEN 'Inactive' THEN 3
        ELSE 4
```

```
END,
ca.lifetime_revenue DESC;
```

Appendix B - Data Dictionary

Table	Field	Meaning
customers	customer_id	Primary key
customers	signup_date	Customer registration date
products	unit_price	Listed product price
products	cost	Product cost
orders	status	completed / returned / cancelled / pending
order_items	quantity	Negative quantity represents return
order_items	discount	Fractional discount
reviews	rating	Valid range 1-5
web_sessions	duration_minutes	Session engagement

Individual contribution statement

This repository was prepared as an individual analytics project. The contributor reviewed the assignment requirements, organized the source files, wrote and validated all ten SQL queries, developed pandas cleaning and integration rules, implemented the four NumPy analytical routines, produced the eight business visualizations, documented business interpretations, and packaged the workflow as a reusable pipeline. The contributor also prepared the report, executive summary, README, processed datasets, audit tables, and GitHub-ready folder structure. Automated generation was used to improve consistency, but all transformations, assumptions, outputs, and conclusions were checked against the supplied data dictionary and project requirements.